

ELASTIC STACK BASED SOC

Centralized Security Monitoring & Threat Detection System

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1. Introduction

1.1 Background

Cybersecurity threats are continuously increasing in both complexity and frequency. Organizations require centralized monitoring systems capable of collecting, analyzing, and detecting suspicious activities in real time. Security Operations Centers (SOC) are responsible for monitoring and responding to security incidents using SIEM (Security Information and Event Management) solutions.

This project implements a Mini SOC Architecture using the Elastic Stack to collect logs from Windows endpoints, analyze events, detect malicious behavior, and generate alerts for security monitoring.

1.2 Problem Statement

Traditional systems generate logs in different locations without centralized monitoring, making incident detection difficult. Security teams require a centralized solution capable of:

- Collecting logs from endpoints
 - Monitoring security events
 - Detecting attacks in real time
 - Visualizing threats through dashboards
 - Supporting incident investigation
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1.3 Project Description

This project implements a Mini Security Operations Center (SOC) using the Elastic Stack to monitor and analyze security events from a Windows endpoint machine. The system collects logs, detects suspicious activities, and visualizes security data through Kibana dashboards.

2. SOC Overview

2.1 What is SOC?

A Security Operations Center (SOC) is a centralized cybersecurity unit responsible for monitoring, detecting, analyzing, and responding to security incidents.

The SOC continuously analyzes logs, alerts, and network activities to identify potential threats and protect organizational infrastructure.

2.2 SOC Workflow

The SOC workflow includes:

1. Data Collection
 2. Log Processing
 3. Threat Detection
 4. Alert Generation
 5. Investigation
 6. Incident Response
 7. Reporting
-

2.3 SOC Importance

A Security Operations Center (SOC) is important for monitoring, detecting, and responding to cyber threats in real time. It helps improve system security by providing centralized visibility of security events and faster incident response. In this project, the SOC environment uses Elastic Stack to analyze logs and detect suspicious activities based on MITRE ATT&CK techniques.

3. Project Overview

3.1 Project Details

The Mini SOC project is designed to simulate a real Security Operations Center environment using Elastic Stack components. The project monitors endpoint activities from a Windows machine and analyzes logs for detecting security threats such as brute force attempts, PowerShell execution, and USB device activity.

The system uses:

- Elasticsearch for log storage and indexing
- Logstash for log processing and forwarding
- Kibana for visualization and monitoring
- APM Server for performance monitoring
- Windows Agent for collecting endpoint logs

The project demonstrates how SIEM technologies can improve security monitoring, threat detection, and incident analysis within an organization.

3.2 SIEM Overview

Security Information and Event Management (SIEM) is a security solution used to collect, analyze, monitor, and correlate logs from different systems and devices in real time. SIEM helps security teams detect suspicious activities, investigate incidents, and respond to threats efficiently.

In this project, the Elastic Stack acts as the SIEM platform by gathering logs from the Windows agent through Logstash, storing them in Elasticsearch, and visualizing security events using Kibana dashboards.

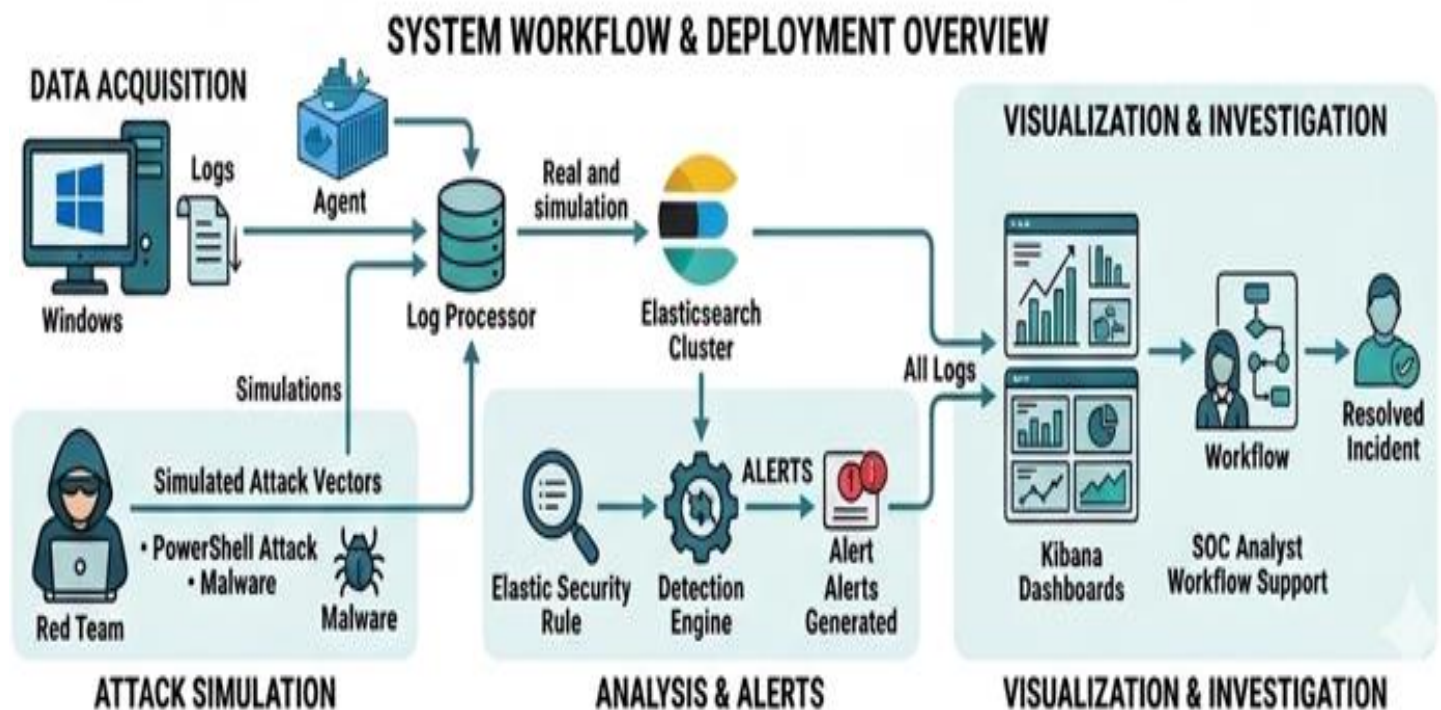
Main Functions of SIEM:

- Centralized log collection
- Real-time monitoring
- Threat detection and alerting
- Event correlation
- Security analysis and reporting

4. Project Objective

The main objectives of this project are:

- Build a centralized log management system
- Monitor Windows security events in real time
- Simulate cybersecurity attacks
- Detect malicious activities using Elastic Security rules
- Generate alerts for suspicious events
- Visualize logs and alerts using Kibana dashboards
- Support SOC analyst investigation workflow



5. Technologies Used

- Elasticsearch: Stores, indexes, and searches large amounts of log and security data quickly and efficiently.
- Kibana: Provides dashboards and visual tools to monitor, analyze, and investigate collected data.
- Elastic Agent: Collects logs, system events, and security data from endpoints and sends them to Elasticsearch.
- Fleet Server: Centrally manages Elastic Agents, configurations, and security policies across devices.
- Elastic Security: Detects threats, generates alerts, and helps analysts investigate suspicious activities.
- Docker: Runs and manages services inside lightweight containers for easy deployment and scalability.
- Windows Endpoint: Acts as the target machine where attacks and security monitoring are simulated.
- PowerShell: Used to execute scripts, simulate attacks, and perform security testing on Windows systems.

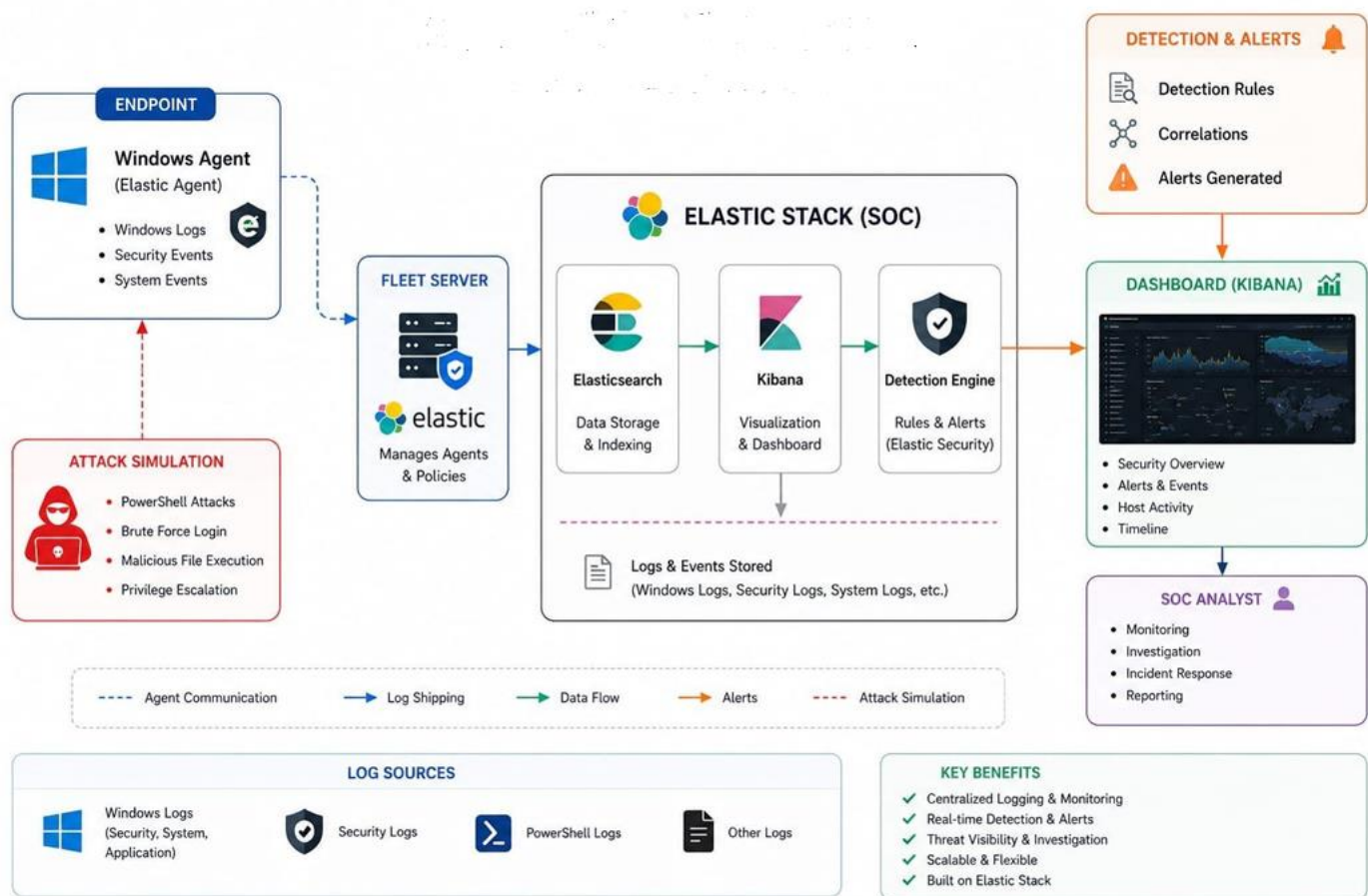
TECHNOLOGY		PURPOSE
	Elasticsearch	Data storage and indexing
	Kibana	Dashboards and visualization
	Elastic Agent	Endpoint log collection
	Fleet Server	Agent and policy management
	Elastic Security	Detection rules and alerts
	Docker	Containerized deployment
	Windows Endpoint	Attack simulation environment
	PowerShell	Attack execution and testing

6. System Architecture

6.1 Architecture Overview

The SOC architecture consists of the following components:

1. Windows Endpoint
2. Elastic Agent
3. Fleet Server
4. Elasticsearch
5. Kibana
6. Detection Engine
7. SOC Analyst Dashboard



6.2 Endpoint Layer

The Endpoint Layer represents the target Windows machine being monitored within the SIEM environment. It acts as the source of system activities, security events, and attack simulations. The Windows endpoint contains the Elastic Agent responsible for collecting:

- Windows Logs
- Security Events
- System Events
- PowerShell Logs

Endpoint IP Address:

10.134.175.222

6.3 Fleet Server

Fleet Server is a central management component in the Elastic Stack that controls and coordinates all connected Elastic Agents. It acts as the communication bridge between endpoints and the Elastic environment, ensuring agents are properly configured and updated.

Responsibilities include:

- Agent enrollment
 - Policy management
 - Configuration deployment
 - Communication management
-

6.4 Elasticsearch

Elasticsearch is the core search and analytics engine in the SIEM environment. It stores and indexes collected logs, allowing fast searching, event analysis, and correlation of security data for threat detection and monitoring.

Elasticsearch is responsible for:

- Log storage
- Data indexing
- Fast searching
- Event correlation

Server Address:

<http://localhost:9200>

6.5 Kibana

Kibana is the visualization and monitoring interface of the Elastic Stack. It allows security analysts to view dashboards, investigate alerts, perform threat hunting, and analyze security events in real time.

Kibana provides:

- Security dashboards
- Alert visualization
- Threat hunting
- Timeline analysis

Server Address:

<http://localhost:5601>

6.6 Logstash

Logstash is a data processing pipeline within the Elastic Stack that collects, filters, and forwards logs from different sources to Elasticsearch. It helps organize and prepare security data for analysis and monitoring.

Logstash is responsible for processing and forwarding logs.

Server Address:

<http://localhost:5044>

6.7 APM Server

APM Server (Application Performance Monitoring Server) is responsible for collecting application performance metrics and telemetry data. It helps monitor application behavior, detect performance issues, and analyze system activity in real time.

APM Server monitors application performance and telemetry data.

Server Address:

<http://localhost:8200>

7. Environment Setup

7.1 Hardware Requirements

Component	Requirement
RAM	Minimum 8 GB
CPU	4 Cores
Storage	50 GB
Network	Internet Connectivity

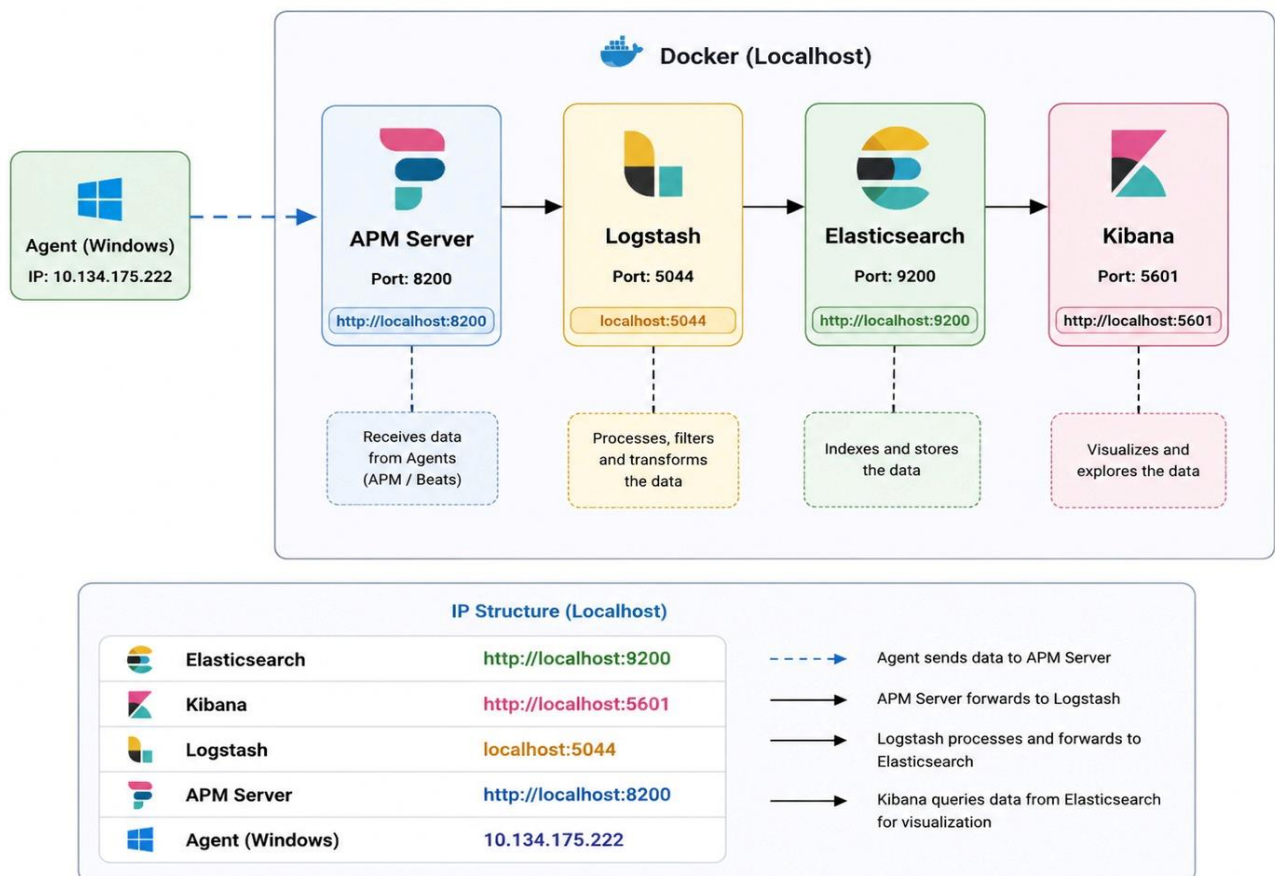
7.2 Software Requirements

Software	Version
Docker	Latest
Docker Compose	Latest
Elastic Stack	8.x
Windows OS	Windows 10/11

7.3 Network Configuration

Service	Address
Elasticsearch	http://localhost:9200
Kibana	http://localhost:5601
Logstash	http://localhost:5044
APM Server	http://localhost:8200
Windows Agent	10.134.175.222

ELK Stack Architecture (Localhost)



8. Elastic Stack Components

8.1 Elasticsearch

Elasticsearch is a distributed search and analytics engine used to store, index, and analyze logs in real time. It enables fast searching and event correlation across large amounts of security data.

Main Features:

- Fast Searching
 - Scalability
 - Real-Time Indexing
 - Distributed architecture
-

8.2 Kibana

Kibana is the visualization platform of the Elastic Stack used for monitoring, analyzing, and investigating security events through dashboards and graphical interfaces.

Main Features:

- Dashboard creation
 - Threat monitoring
 - Security analysis
 - Alert investigation
-

8.3 Fleet Server

Fleet Server is the centralized management component responsible for controlling Elastic Agents, deploying configurations, and managing communication between agents and the Elastic environment.

8.4 Elastic Agent

Elastic Agent is responsible for collecting logs and telemetry data from endpoints and forwarding them to the Elastic Stack for monitoring and analysis.

Collected Logs:

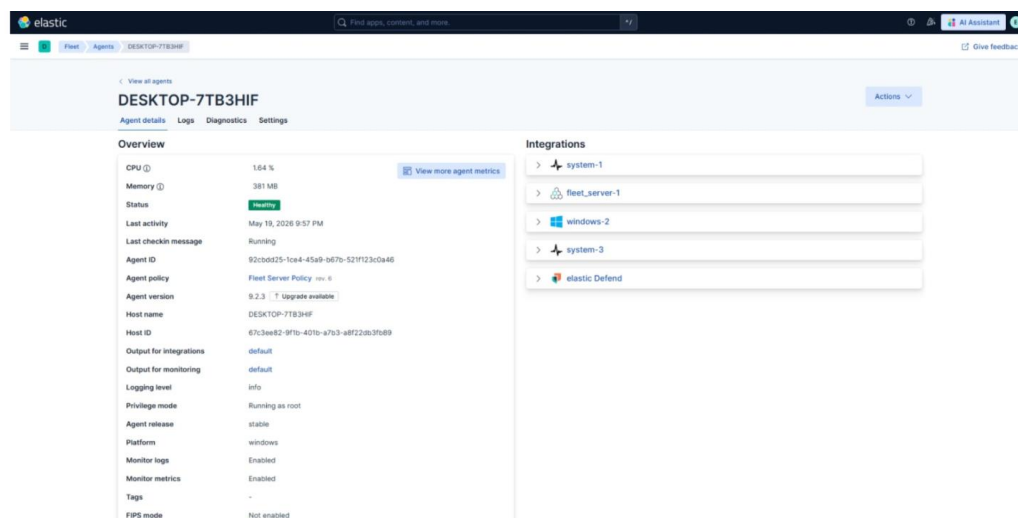
- Windows Security Logs
- System Logs
- PowerShell Logs
- Application Logs

8.5 Elastic Security

Elastic Security is the SIEM and security analytics solution within the Elastic Stack used for detecting threats, generating alerts, and supporting threat hunting operations.

Main Features:

- Detection rules
- Alert generation
- Threat hunting
- MITRE ATT&CK mapping



9. Installation & Configuration

9.1 Docker Installation

Install Docker and Docker Compose on the host machine.

Verification Commands:

```
docker --version
```

```
docker compose version
```

9.2 Deploy Elastic Stack

Run the Elastic Stack containers using Docker Compose.

Example Command:

```
docker compose up -d
```

Verify Running Containers:

```
docker ps
```

9.3 Access Kibana

Open Kibana in the browser:

<http://localhost:5601>

9.4 Configure Fleet Server

Steps:

1. Open Kibana
2. Navigate to Fleet Management
3. Create Agent Policy
4. Add Windows Integration
5. Generate Enrollment Token

9.5 Install Elastic Agent on Windows

Steps:

1. Download Elastic Agent
2. Copy enrollment command
3. Run PowerShell as Administrator
4. Execute enrollment command
5. Verify agent connectivity

9.6 Verify Data Collection

Check:

- Fleet Agent Status
- Elasticsearch indices
- Kibana dashboards
- Security alerts

10. Log Collection Workflow

10.1 Log Sources

The system collects logs from:

- Windows Security Logs
 - System Logs
 - PowerShell Logs
 - Application Logs
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10.2 Data Flow

The log collection workflow follows the sequence below:

Windows Endpoint → Elastic Agent → Fleet Server → Elasticsearch →
Kibana → Detection Engine

10.3 Log Monitoring

The SOC analyst monitors:

- Failed Login attempts
- PowerShell activity
- USB device events
- Security alerts
- Host activity

11. Attack Simulation

11.1 Brute Force Attack

MITRE ATT&CK ID: T1110

Description:

A brute force attack attempts multiple password combinations to gain unauthorized access.

Simulation Steps

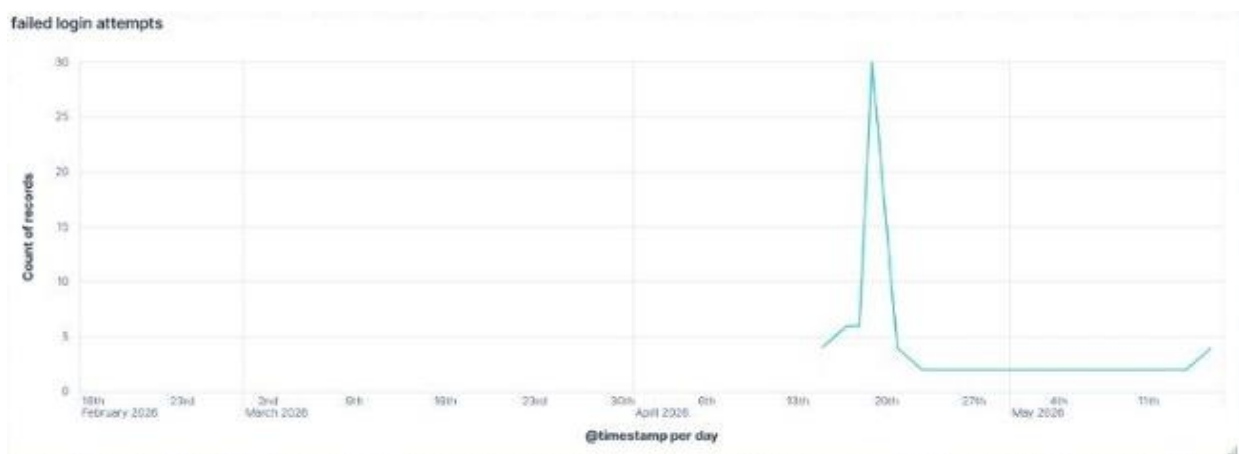
1. Execute multiple failed login attempts
2. Generate Windows authentication logs
3. Trigger detection rule
4. Generate alert in Kibana

Expected Detection

- Multiple failed login events
- High authentication failure count
- Alert generation

Defend Method:

Strengthen authentication using strong passwords, MFA, and account lockout policies, Detect and respond by monitoring failed logins, blocking IPs, and resetting compromised accounts.



11.2 PowerShell Attack

MITRE ATT&CK ID: T1059

Description:

PowerShell attacks involve malicious script execution using PowerShell commands.

Simulation Steps

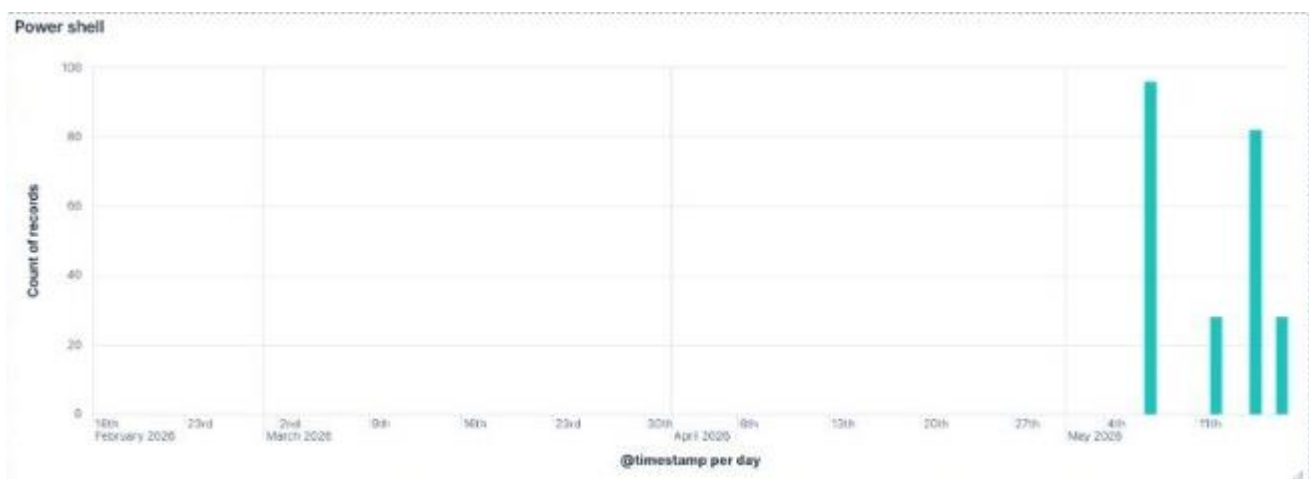
1. Execute suspicious PowerShell commands
2. Generate PowerShell logs
3. Trigger PowerShell detection rule
4. Generate security alert

Expected Detection

- Suspicious PowerShell execution
- Encoded commands
- Script execution monitoring

Defend Method:

Reduce risk by restricting PowerShell execution and enabling full logging and auditing, respond by investigating suspicious commands, isolating systems, and blocking malicious scripts.



11.3 USB Data Transfer Attack

MITRE ATT&CK ID: T1052

Description:

USB attacks involve unauthorized data transfer using removable storage devices.

Simulation Steps

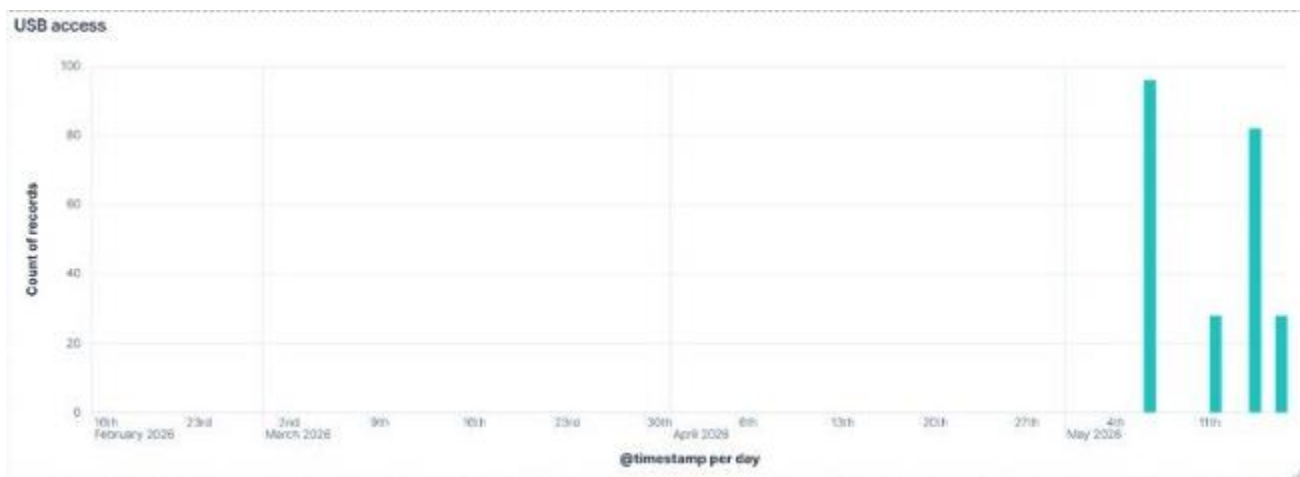
1. Connect USB device
2. Transfer files
3. Generate device activity logs
4. Trigger USB detection rule

Expected Detection

- USB device insertion events
- File transfer monitoring
- Alert generation

Defend Method:

Control USB access through device policies, restrictions, and file scanning. Detect via monitoring USB activity, reviewing logs, and blocking unauthorized data transfers.



11.4 Windows Reconnaissance Commands Attack

MITRE ATT&CK ID: T1082 / T1057

Description:

Reconnaissance commands are used by attackers to collect information about a system, users, processes, and network settings before launching further attacks. They help identify the environment and possible weaknesses.

Simulation Steps

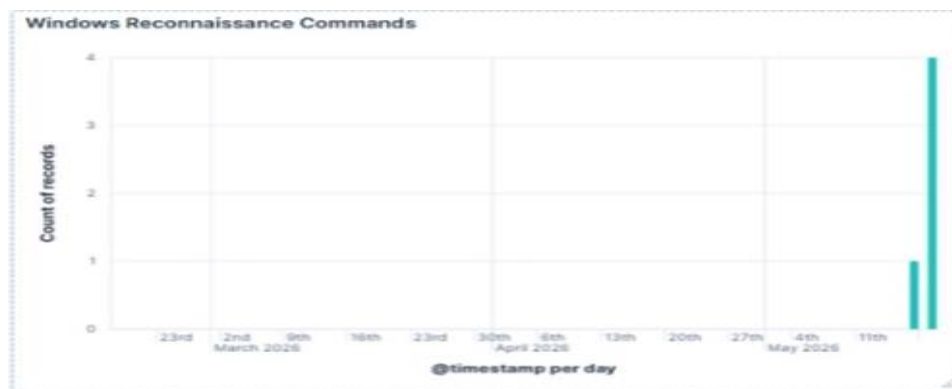
1. Execute Windows reconnaissance commands
2. Generate PowerShell or CMD activity logs
3. Collect command execution events through Elastic Agent
4. Trigger reconnaissance detection rule in Elastic Security

Expected Detection

- Suspicious command execution events
- PowerShell reconnaissance activity
- Process monitoring logs
- Alert generation in Kibana dashboards

Defend Method:

Limit exposure by restricting privileges and enabling command-line monitoring and logging, respond by analyzing suspicious commands, blocking tools, and isolating affected endpoints.



12. Detection Rules & MITRE Mapping

12.1 Detection Rules Overview

Detection rules are used to identify suspicious activities based on event patterns and behaviors.

12.2 MITRE ATT&CK Mapping

Attack	MITRE Technique
Brute Force	T1110
PowerShell	T1059
USB Data Transfer	T1052
Windows Reconnaissance Commands	T1082 / T1057

12.3 Alert Workflow

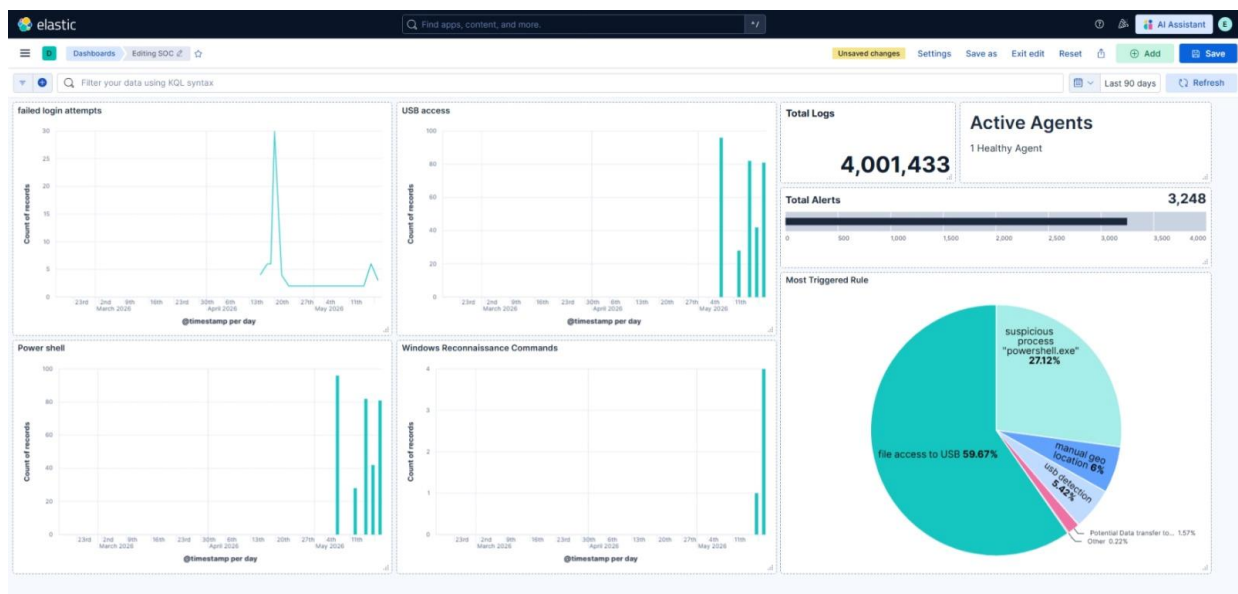
1. Event Generated
2. Rule Matching
3. Alert Creation
4. Alert Visualization
5. Analyst Investigation

13. Dashboards & Monitoring

13.1 Kibana Security Dashboard

The Kibana dashboard provides:

- Security Overview
- Alert Monitoring
- Event Visualization
- Host Monitoring
- Timeline Investigation



13.2 Security Monitoring

The SOC analyst continuously monitors:

- Authentication failures
- Suspicious processes
- PowerShell activity
- Device events
- Security alerts

14. Incident Investigation Workflow

Investigation Process

The SOC analyst follows the workflow below:

1. Alert Received
 2. Event Review
 3. Timeline Analysis
 4. Threat Validation
 5. Root Cause Analysis
 6. Incident Response
 7. Documentation
-

15. Challenges & Solutions

Challenge	Solution
Elastic Agent connectivity issues	Reconfigured Fleet policies
Missing logs in Kibana	Verified integrations and indices
Detection rule false positives	Tuned detection thresholds
Docker networking issues	Reconfigured container networking

16. Results

The project successfully achieved:

- Centralized log collection
- Real-time monitoring
- Attack detection
- Alert generation
- Dashboard visualization
- Incident investigation support

Key achievements include:

- Detection of brute force attacks
 - Detection of malicious PowerShell activity
 - Monitoring USB device activities
 - Real-time security dashboards
-

17. Future Improvements

Future enhancements may include:

- Linux endpoint integration
- Threat intelligence feeds
- Email notifications
- Machine learning anomaly detection
- SOAR integration
- Cloud monitoring
- Network traffic analysis

18. Conclusion

This project demonstrates the implementation of a Mini SOC architecture using the Elastic Stack for centralized monitoring and threat detection.

The system successfully collects logs, detects malicious activities, generates alerts, and supports incident investigation through Kibana dashboards and Elastic Security.

The project highlights the importance of SIEM solutions in improving cybersecurity visibility and incident response capabilities.

19. References

1. Elastic Official Documentation
2. MITRE ATT&CK Framework
3. Microsoft Windows Event Logging Documentation
4. Docker Official Documentation
5. Elastic Security Documentation